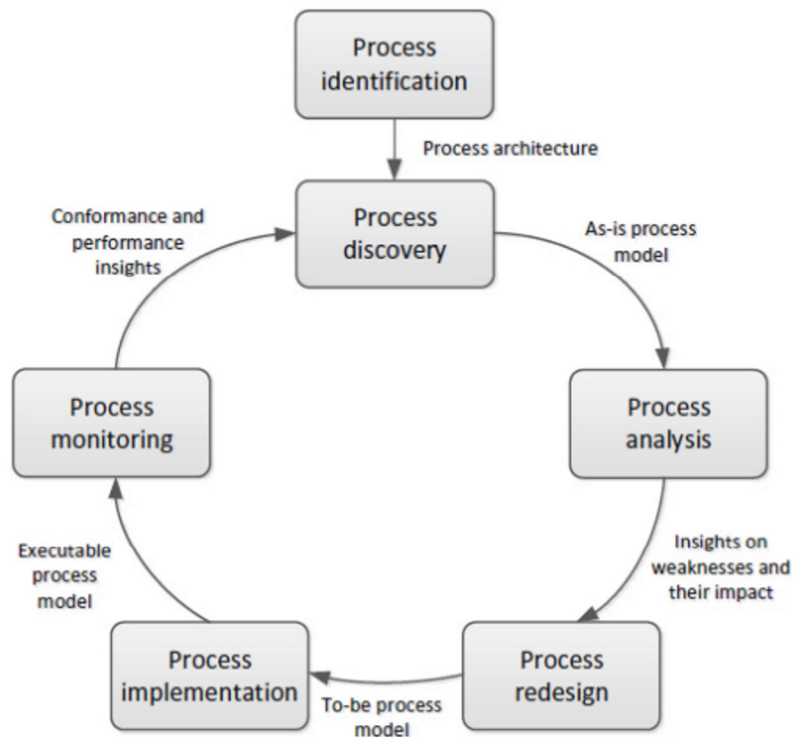


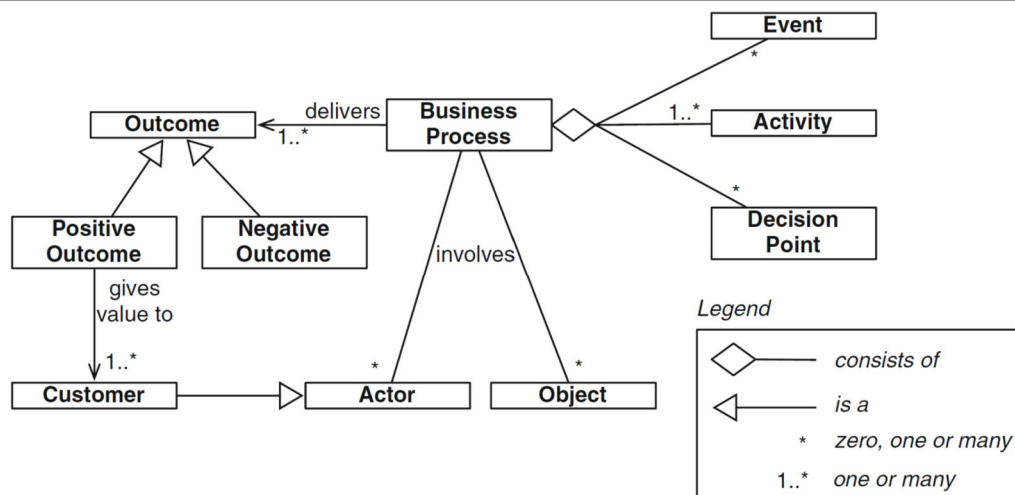
# Das Wichtigste

Freitag, 9. August 2024 13:09



## Business Processes

|                              |                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------|
| <u>Actors:</u>               | Entities that are performing process activities and generate events               |
| <u>Tasks and Activities:</u> | Units of work, Time-consuming, Resource-demanding, State-changing                 |
| <u>Events:</u>               | Passiv elements, Stands for conditions or circumstances, Atomic and instantaneous |
| <u>Decision Points:</u>      | Effect future process execution                                                   |
| <u>Physical Objects:</u>     | Can undergo state changes, Contain physical or electronic information             |
| <u>Others:</u>               | Informational Object, electronic documents, electronic records                    |
| <u>Outcome</u>               | Positiv Negativ                                                                   |





## Prozess Identification

### Definition Phase

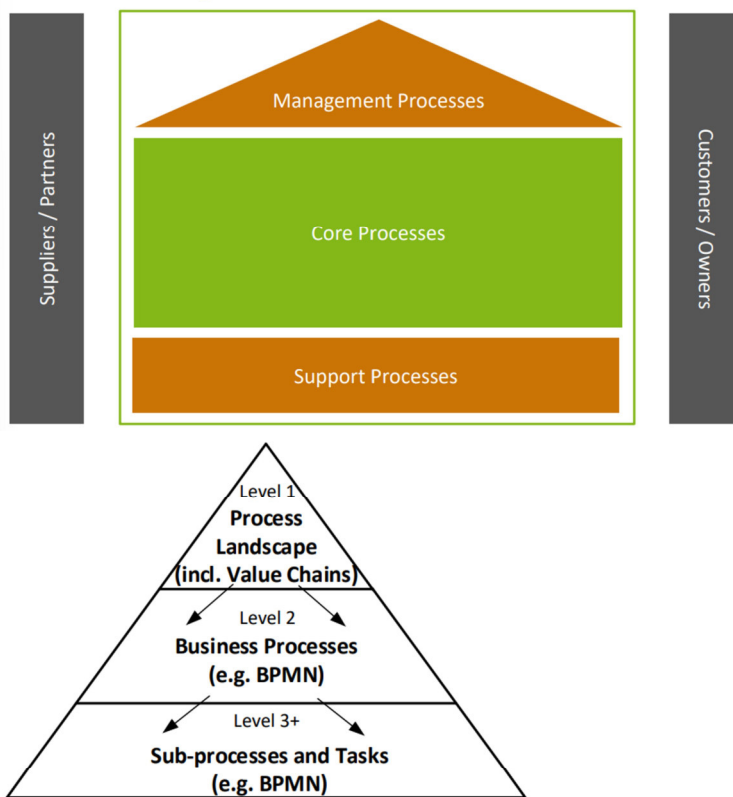
- Understand
- Define
- Determine

### Selection Phase

- Define criteria
- Prioritize
- Select

## Definition

### Process Categories



## Building a Process Landscape Model (Level 1) - Overview

1. Clarify terminology
2. Identify end-to-end processes
3. Identify ist sequential processes
4. Identify ist major management and support processes
5. Decompose and specialize
6. Compile process profile
7. Check completeness and consistency

## Selection

### Criteria for Priortization

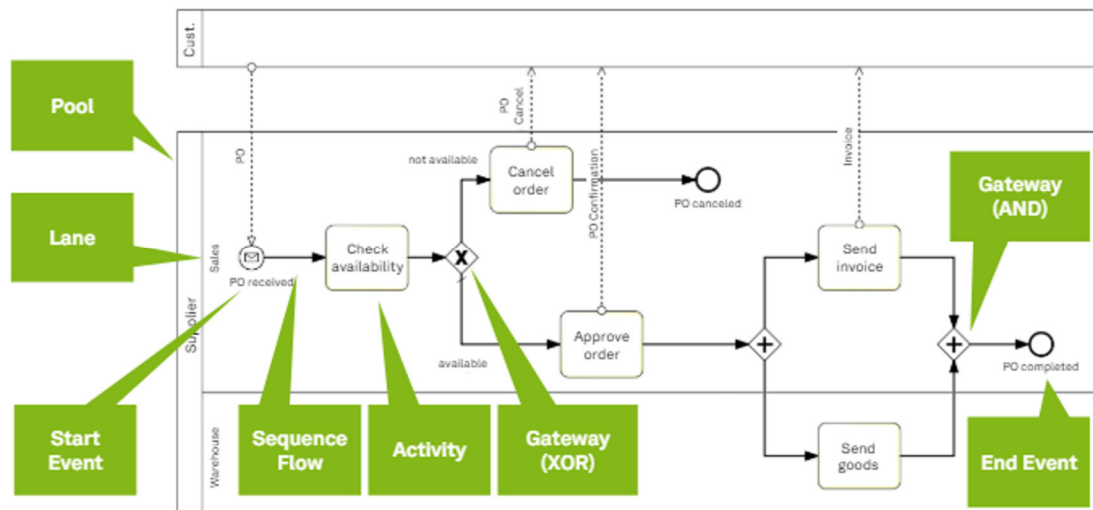
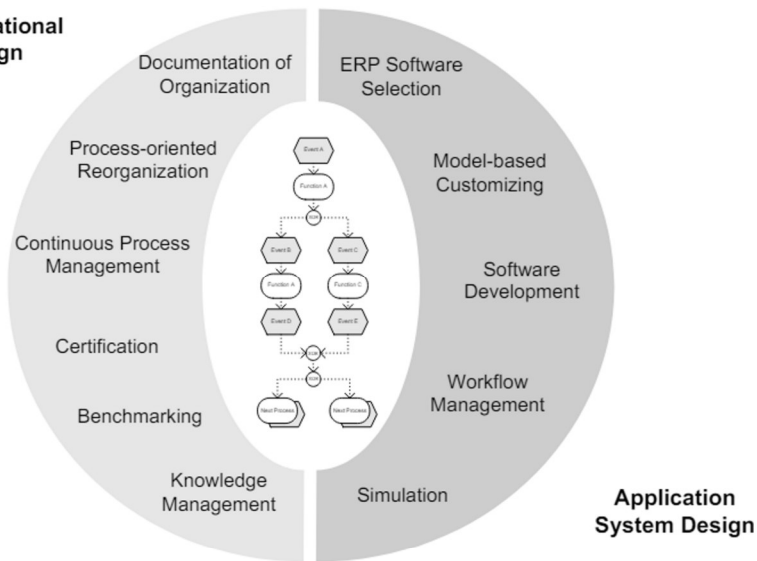
- Strategic Importance
- Health (Dysfunction)
- Feasibility

Result: Portfolio

## Modeling

- Mapping
- Abstraction
- Pragmatic purpose

## Organizational Design



## Process Discovery

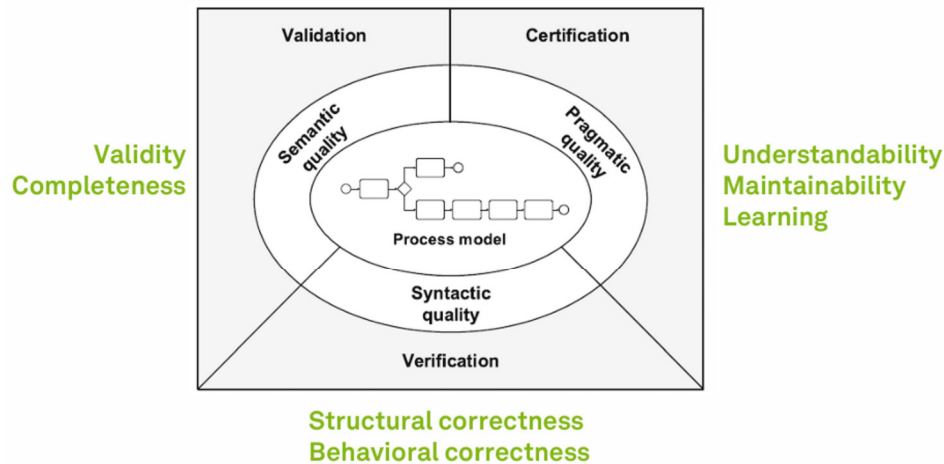
### Phases

1. Define the setting
  - Actors
  - Challenges
  - Problems
  - Guidelines
2. Gather information
  - Evidence-based
    - Document Analysis
    - Observation
    - Automated process discovery
  - dialog-based
    - Interview
    - Workshop

| Aspects     | Evidence | Interview   | Workshop    |
|-------------|----------|-------------|-------------|
| Objectivity | high     | medium-high | medium-high |

|                       |            |        |        |
|-----------------------|------------|--------|--------|
| Richness              | medium     | high   | high   |
| Time consumption      | low-medium | medium | medium |
| Immediacy of feedback | low        | high   | low    |

3. Conduct the modeling task
  1. Identify the process boundaries
  2. Identify the activities and events
  3. Identify resources and their handoffs
  4. Identify the control flow
  5. Identify additional elements
4. Assure process model quality



#### Sound Model

- Option to complete
- Proper completion
- No dead activities

### **Process Modeling Guidelines and Conventions**

#### Objectives

- Improve model consistency and standardization
- Reduce the dependency
- Facilitate access

#### Restrictions

- Vocabulary (avoid certain elements)
- Structure (limit the structure of the model)
- Semantics (avoid element meanings)
- Appearance (restrict the model appearance)

#### 7PMG

- G1) Use as few elements in the model as possible
- G2) Minimize the routing paths per element
- G3) Use one start and one end event
- G4) Model as structured as possible
- G5) Avoid, where possible, Or routing elements
- G6) Use verb-object activity labels
- G7) Decompose a model more than 30 elements

### **Selection of Process Analysis Techniques**

#### Qualitative Analysis

- Value-added Analysis
- Waste Analysis
- Stakeholder Analysis
- Root cause Analysis

### Quantitative Analysis

- Flow Analysis
- Queueing Analysis
- Process Simulation

### Qualitativ

#### **Value-added Analysis**

- Classify each step into
  - Value-adding (VA): value or satisfaction to the customer
  - Business value-adding (BVA): Necessary or useful for the business
  - Non-value-adding (NVA): Everything else including handovers, delays, and rework

#### Guidelines for Elimination

- Minimize and Elimination NVA steps
- Minimizing and Eliminating BVA steps is a trade-off

#### **Waste Analysis**

- Move
  - Transportation (send, receive)
  - Motion (drop-off, pick-up, go to)
- Hold
  - Inventory (large work-in-progress)
  - Waiting (idle time between tasks)
- Overdo
  - Defects (compensation to fix defects)
  - Overprocessing (performing what is not needed)
  - Overproduction (unnecessary cases)
  - (Skills (resource underutilization, idle resources))

#### **Stakeholder Analysis**

- Customers
- Process participants
- Process owner
- External parties
- Sponsor

#### **Root Cause Analysis**

- causal factors (root cause)
- Contributing factors (enabler, multiplier)
- Interviews or workshops
- Results are documented as
  - Cause-Effect diagrams
  - Why-why diagrams

#### 6 M's

- Machine (technology)
- Method (process)
- Material
- Man
- Measurement
- Milieu

#### Why-why Diagramm

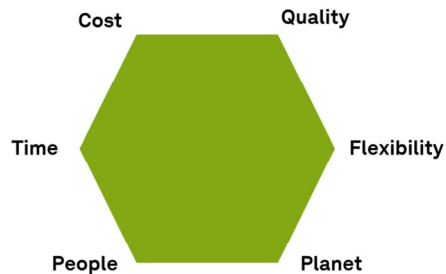
- Ask 5 times --> root-cause
- Focus on Level 3 & 4
- Leaves should be fundamental factors (atomic, cannot be explained otherwise)

### Quantitativ

## The Devils Quadrangle



## Tripple Bottom Line



### Time

- Cycle time / throughput time
- Processing time
- Waiting time

### Cost

- Fixed
- Variable

### Quality

- Internal Quality
  - As perceived by internal participants
- External Quality
  - As perceived by the client/customer

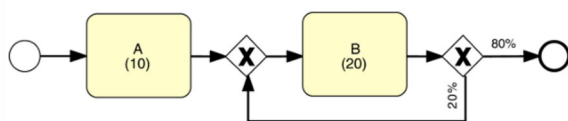
### Flexibility

- Reaction to Changes
  - Resources
  - Process
  - Management
  - Organization
- Run time vs. Build time

## Flow Analysis

- Time, error rates, requirements and estimation of cost

1. Performance of a Sequence
2. Performance of a XOR Block
3. Performance of a AND Block
4. Performance of a Loop Block (Rework)



$$CT = \frac{T}{1 - r}$$

$r$  = rework probability

$T_i$  = cycle time of a process fragment

$CT$  = cycle time of a process

### Cycle Time Efficiency

- Ratio of theoretical processing time relative to overall processing time
- Theoretical cycle time does not have waiting / idle time

$TCT$  = theoretical cycle time ↔

$CT$  = cycle time of a process ↔

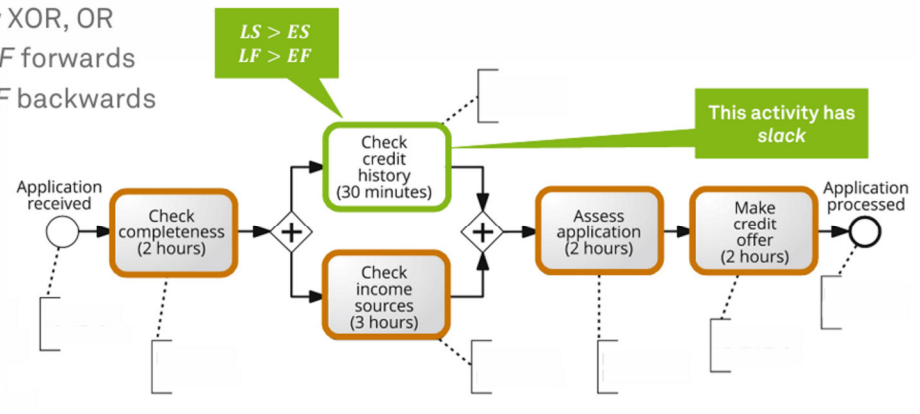
$CTE$  = cycle time efficiency

$$CTE = \frac{TCT}{CT}$$

### Critical Path Method

- Identify tasks that contribute to the theoretical cycle time
  - Remove or simplify XOR, OR
  - Calculate  $ES$  and  $EF$  forwards
  - Calculate  $LS$  and  $LF$  backwards

$ES$  = early start  
 $EF$  = early finish  
 $LS$  = late start  
 $LF$  = late finish



### Little Laws for What-if Analysis

- Average cycle time of a process
- Average number of new process instances per time unit (arrival rate)
- Average number of process instances active at a given time (work-in progress, started but not completed)
  - If cycle time or arrival rate increases, WIP increases, and the process slows down
  - If the arrival rate increases, the cycle time must decrease or the WIP increases as well

$$WIP = \lambda \times CT$$

$CT$  = cycle time of a process

$\lambda$  = arrival rate

$WIP$  = work-in-process

Holds for any stable process

Arrival rate can be substituted by throughput  
 (average completed instances per time unit)

### Capacity and Bottlenecks

- Theroretical capacity
  - Maximum number of instances that can be completed per time unit given a set of resources

$$\mu_p = \frac{uc}{ul}$$

$p$  = resource pool  
 $uc$  = unit capacity  
 $ul$  = unit load

- Resource utilization (occupation rate)
  - Share of working vs. Idle workers

$$\rho_p = \frac{\lambda}{\mu_p}$$

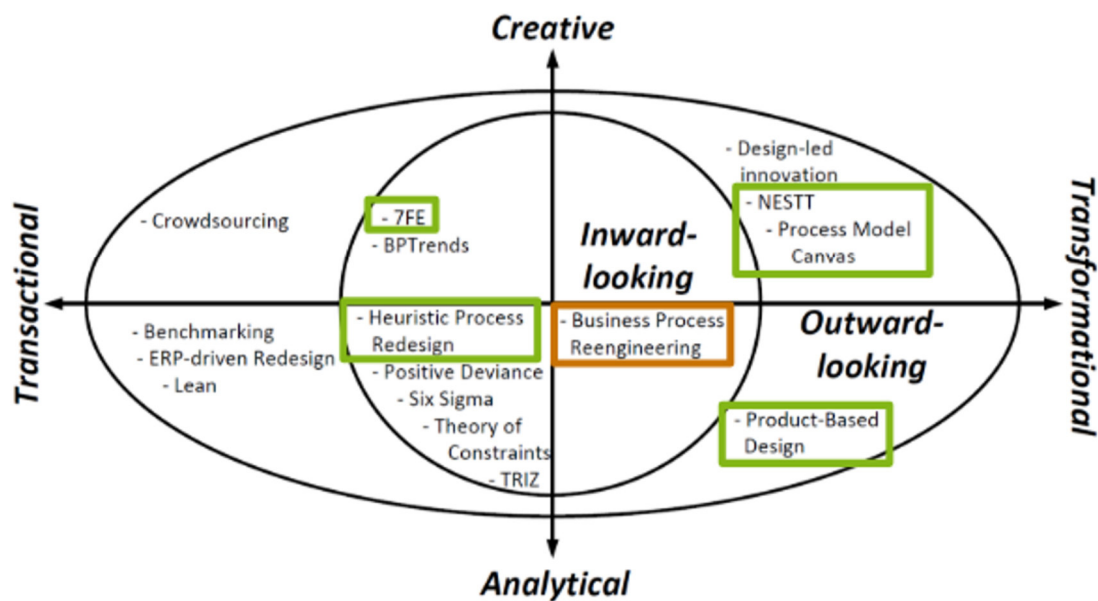
$\lambda = \text{arrival rate}$

- First poll reaches 100% resources is the bottleneck and determines overall theoretical capacity

### Process Simulation

1. Model the process
2. Enhance the process model with simulation information
3. Run Simulation
4. Analyze
5. Repeat
6. Cross-check

### Process Redesign Orbit



### 7FE

- Framework for projects or programs
- Methods: workshops

### Stages of Innovative Phase

1. Prepare: Collect inputs for workshops
2. Generate: Ideas with facilitators, converge to consensus and determine scenarios
3. Validate: Test effectiveness and feasibility

### Heuristic Process Redesign Procedure

- Uses a list of 29 redesign heuristics to determine improvement actions

### Stages of Heuristic Redesign

1. Initiate
2. Design
3. Evaluate
4. Execute in highly iterative and overlapping ways



| Dimension   | Heuristic            | Time | Cost | Quality | Flex. |
|-------------|----------------------|------|------|---------|-------|
| Time        | Parallelism          | +    |      |         |       |
|             | Case-based work      | +    |      |         |       |
| Cost        | Activity elimination | +    | +    |         |       |
|             | Empower              | +    | +    |         | +     |
| Quality     | Triage               |      |      | +       |       |
|             | Case assignment      | +    |      | +       |       |
| Flexibility | Flexible assignment  | +    |      | +       | +     |
|             | Centralization       | +    |      |         | +     |

## NESTT

- Stages
  - Navigate, Expand, Strengthen, Tune/Take-off
- Workshop setting
  - 8 to 10 people use the walls and the floor of the room to visualize and address different viewpoints

## Business Process Reengineering

- Clean slate approach
- Information technology (Automation)
- Product (or service) is central concepts rather than process

## Product-based Design

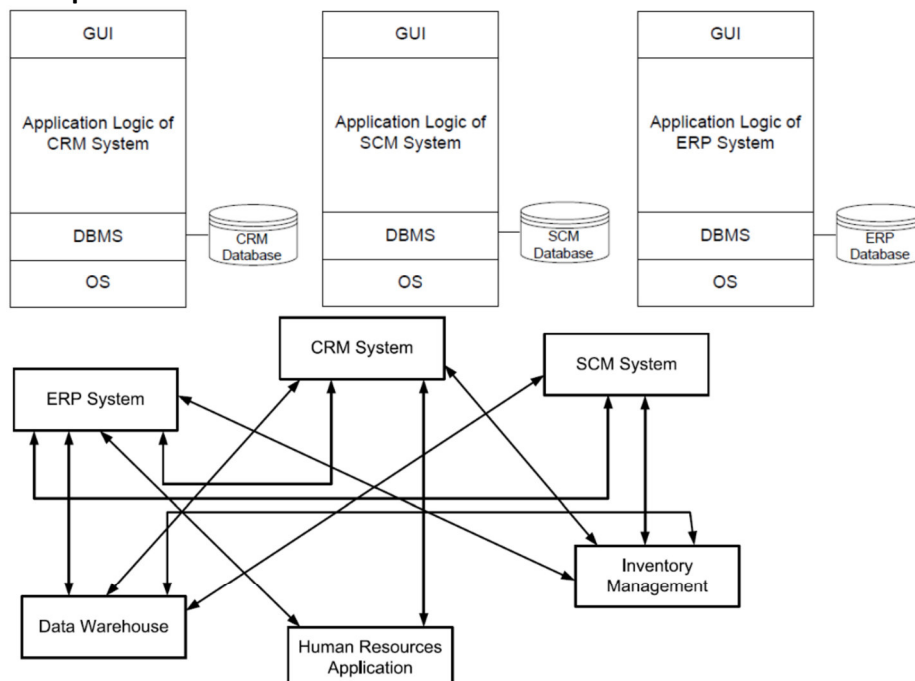
- leanest, most performative process possible

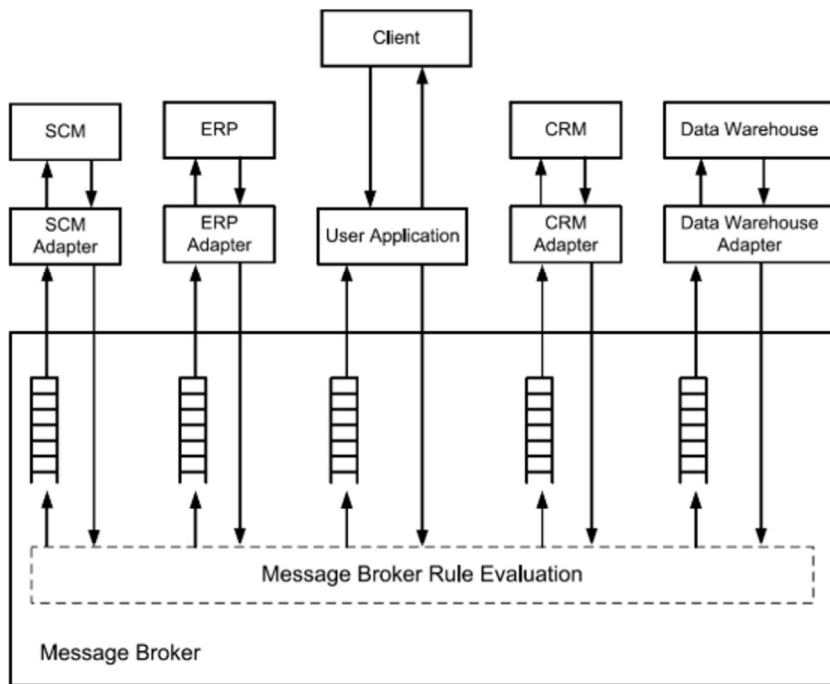
### Product-based Design Process SADEE

1. Scoping:
2. Analysis:
3. Design:
4. Evaluation:
5. Execute in an iterative ways

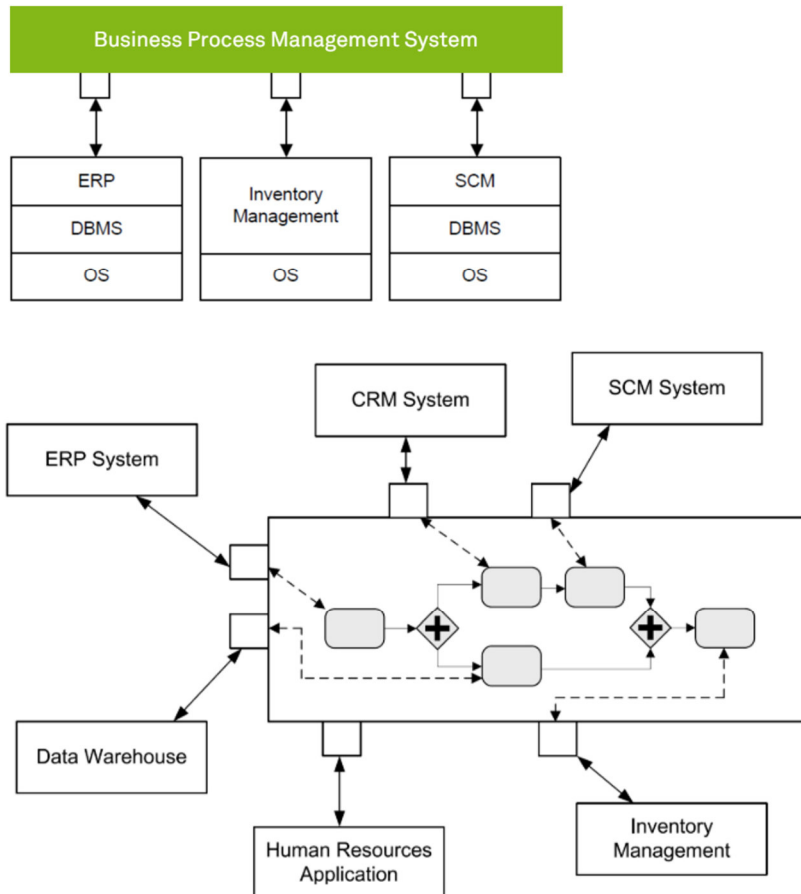
## Technology

### Enterprise Software bevor BPM



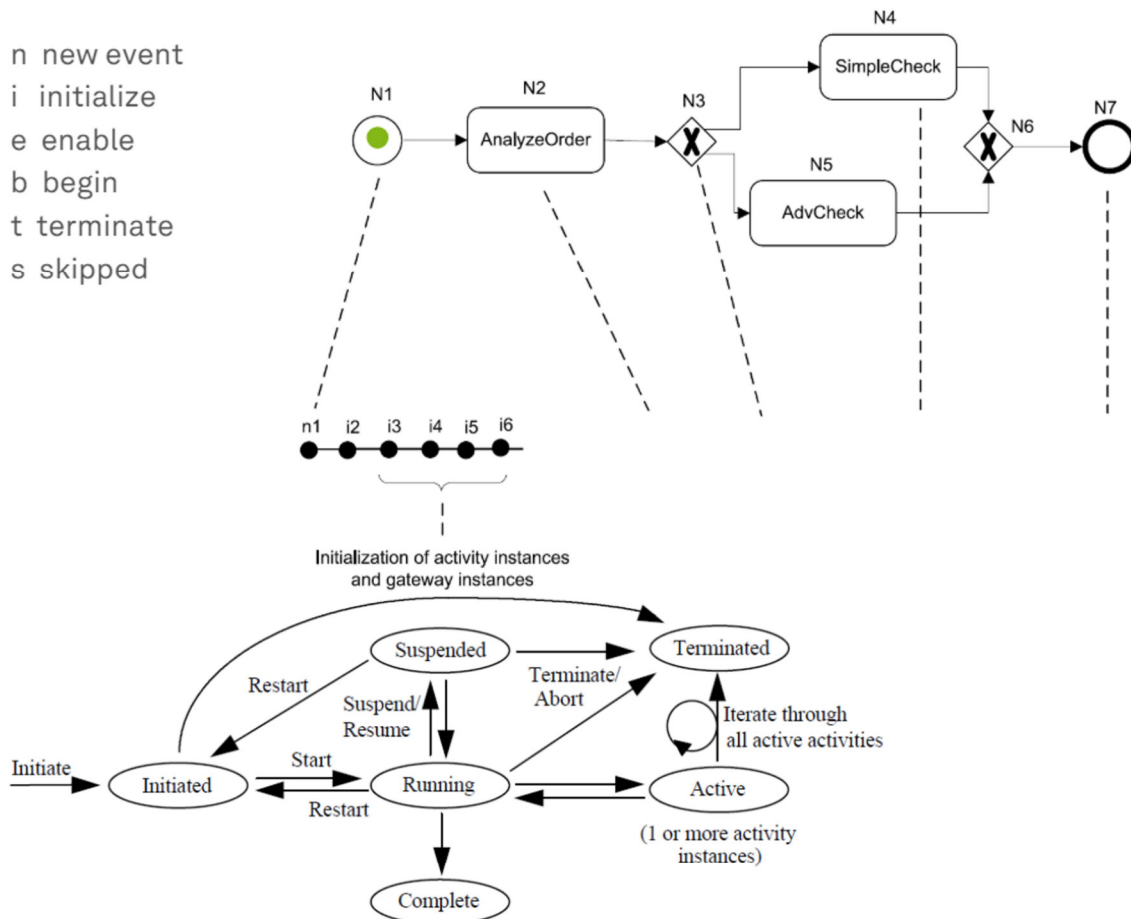


### Process Orchestration with BPMS

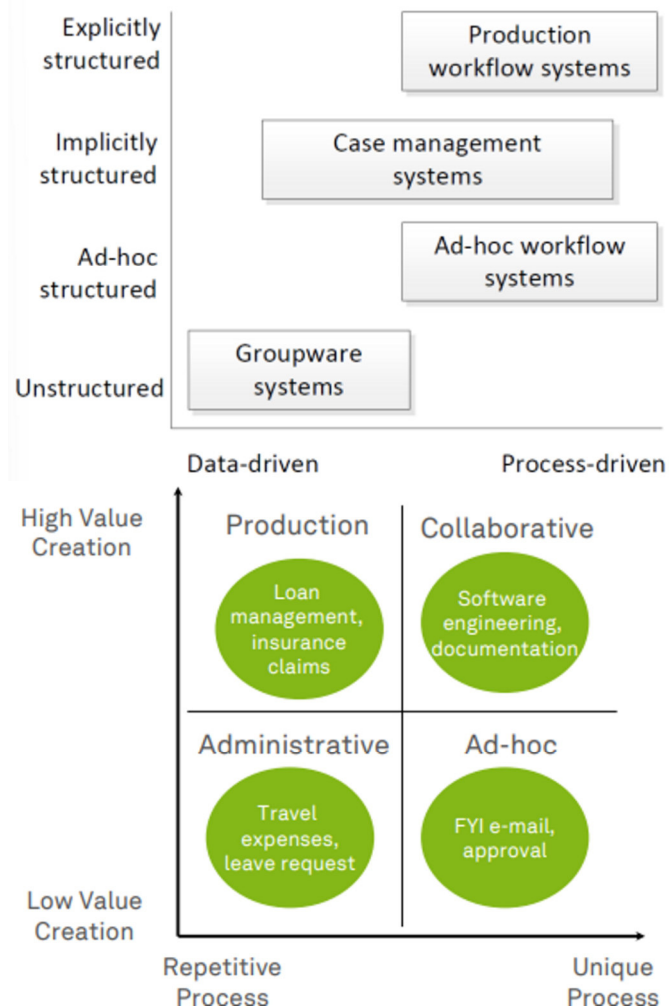


### Stages of Process Instances

n new event  
i initialize  
e enable  
b begin  
t terminate  
s skipped

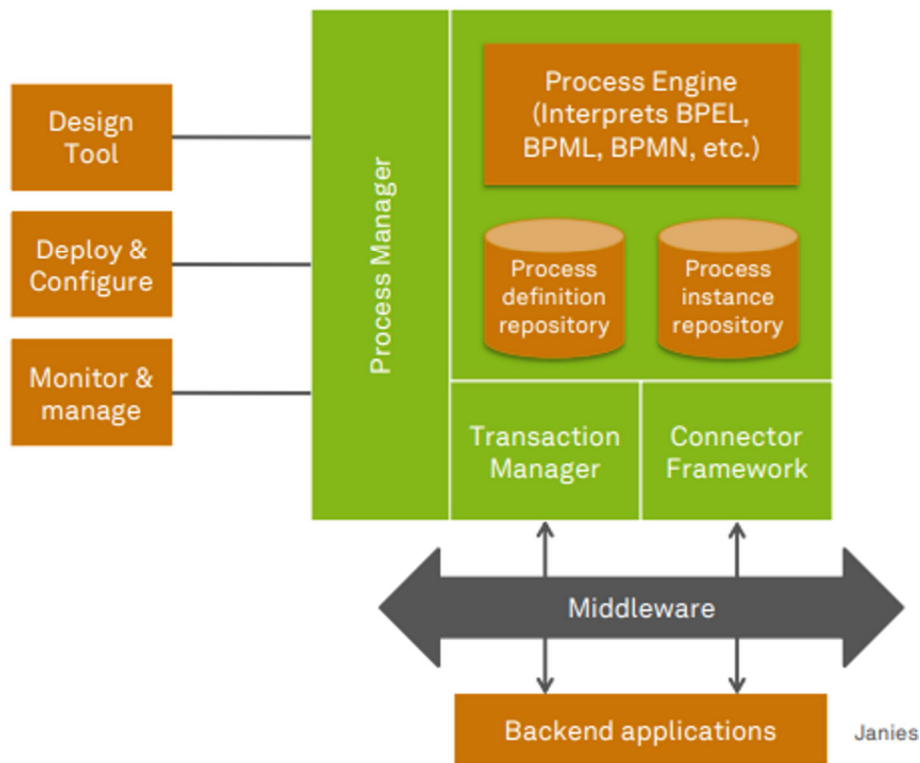


## Types of Business Process Management Systems



## Generic BPMS Architecture

- technical platform for realizing BPM
  - Process engine, executing digital process Models
  - Buildtime tools for process modeling, simulation and deployment
  - Run time facilities for process monitoring and management



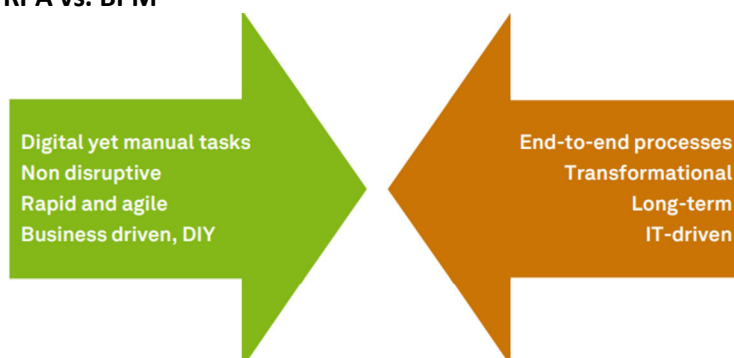
## BPMS Architecture

- BPMS standard components
  - Tooling: Graphically model processes and control BPMS
  - Repository: Store models in a process definition language
  - Process Engine: Execute process in a process engine
- BPMS might support additional functionality like
  - Different modeling languages
  - Simulation and analysis methods
  - Application integration (EAI middleware features)
  - Monitoring and management

## Advantages of a BPMS

- Workload Reduction
- Flexible System Integration
- Execution Transparency
- Rule Enforcement

## RPA vs. BPM



## Process Monitoring

- Use the data generated by the execution of a business process

### Offline Process Monitoring

- Covers historic process executions

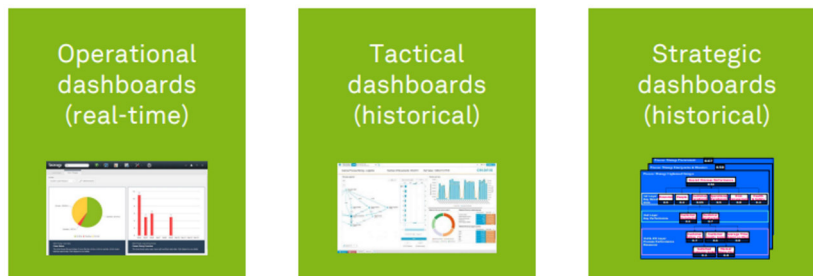
### Online Process Monitoring

- Covers currently running process instances

### Process Monitoring Dimensions

- Focus
- Presentation
- Timeframe
- Aggregation
- Data Scope
- Object
- Process Scope

## Dashboards



## Informational Latency

- Infrastructural Latency
  - Data Latency
  - Analysis Latency
- Decision Latency
- Implementation Latency

## Process Mining

- exploit event data in a meaningful way
- Use event data to discover processes, check compliance, analyze bottleneck, compare process variants, and suggest improvements

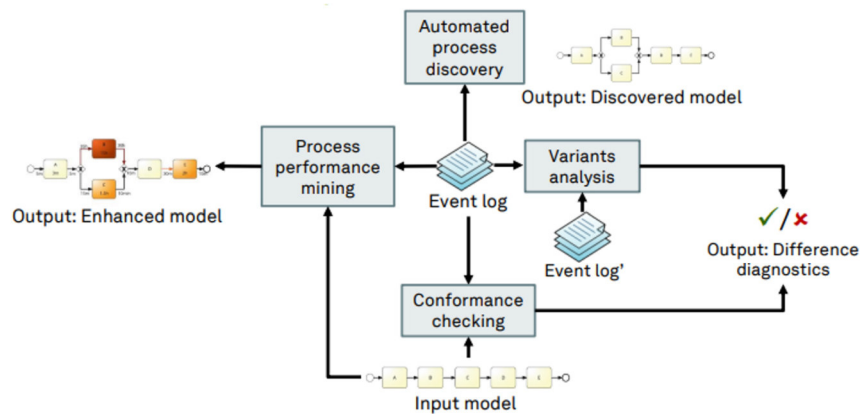
### Challenges

- Correlation:
- Timestamps:
- Longevity:
- Granularity:
- Scoping:

### Structure

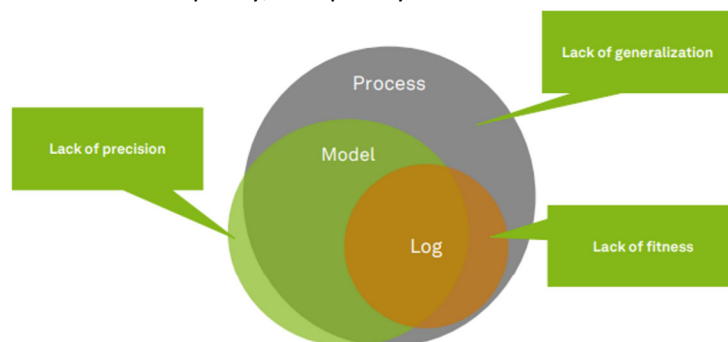
- Process
- Cases
- Events
- Attributes

## Techniques



### Automated Process Discovery

- Construct model that captures behavior of event log
- Dependency Graph
- Abstraction and Filtering
- Discovery Algorithms
- Quality Measures
  - Fitness
  - Precision
  - Generalization
  - Simplicity/Complexity



### Conformance Checking

- Determine whether the execution of a process follows predefined rules or constraints, or if it violates
- Analysis
  - Replay, trace alignment, and behavioral alignment

#### Conformance Checking of Control Flow

- Explicit Constraints
  - Mandatory constraint
  - Exclusiveness constraints
  - Ordering constraints
- Normative process model
  - Unfitting log behavior
  - Additional model behavior

### Variant Analysis

- Determine the root cause for (performance) variation between cases
- Analysis: Differences across subsets of cases in a process

#### Types

- Manual comparison
- Log Delta Analysis
- Associated Rules

## Process Performance Mining

- Goal
  - Quantify and visualize the performance of a process automatically
- Analysis
  - Can be organized around the devil's quadrangle of time, cost, quality, and flexibility
- Result
  - Quantified measures that evaluate a process

### Time

- Dotted chart
  - Timeline chart
- Performance-enhanced dependency graphs

### Cost

- Direct cost of labor
- Overhead cost of labor
- Activity-based Costing allocates indirect cost to an activity's direct cost

### Quality

- not directly visible from event logs

Cycle time of a  
rework block

$$CT = \frac{T}{1 - r}$$

$$r = 1 - \frac{T}{CT}$$

Rework probability

### Flexibility

- compare the desired level of flexibility with the actual flexibility
- A lack of flexibility hints at a lack of standardization
- Suggest Measures

- Distinct
  - Traces DE with unique sequences in event log L
  - Issues with concurrency

$$DE = |L|$$

- Degree of optionality
  - Tasks  $T_{opt}$  in a sequence of tasks T are optional

$$OPT = \frac{T_{opt}}{T}$$

## Enterprise Capability

### Reasons for Failure

- not on business goals
- single source of truth
- isolated silos
- Inability to change

### CMMI Levels of Capability and Performance

| Level   | Continuous Representation<br>Capability Levels | Staged Representation<br>Maturity Levels |
|---------|------------------------------------------------|------------------------------------------|
| Level 0 | Incomplete                                     |                                          |
| Level 1 | Performed                                      | Initial                                  |
| Level 2 | Managed                                        | Managed                                  |
| Level 3 | Defined                                        | Defined                                  |
| Level 4 |                                                | Quantitatively Managed                   |
| Level 5 |                                                | Optimizing                               |

### Level 1: Initial

